



Ascham School

Mathematics Extension 1

Trial HSC Examination

Monday 29th July 2019
2 hours

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using black non-erasable pen.
- NESA-approved calculators may be used.
- A Reference Sheet is provided.
- In Questions 11–14, show relevant mathematical reasoning and/or calculations.

Total marks – 70

Section I Pages 3 – 6

10 marks

- Use the Multiple Choice Answer Sheet provided to answer Q1-10.
- Allow about 15 minutes for this section.

Section II Pages 7 – 13

60 marks

- Answer Questions 11-14.
- Answer each question in a new booklet.
- Label all sections clearly with your name/number and teacher's initials.
- Allow about 1 hour and 45 minutes for this section.

Section I

10 marks

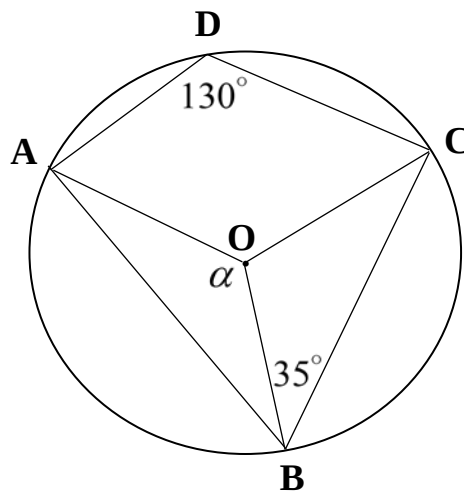
Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 A, B, C and D are points on the circumference of the circle centre O .
 $\angle OBC = 35^\circ$, $\angle ADC = 130^\circ$ and $\angle AOB = \alpha$. The size of α is:

- (A) 70°
- (B) 110°
- (C) 150°
- (D) 185°



NOT TO SCALE

- 2 Consider the function $f(x) = x^2 + 6x$.
Select the largest domain of $f(x)$ for which there exists an inverse function $f^{-1}(x)$.

- (A) $x \geq 0$
- (B) $x \geq -3$
- (C) $x \geq -6$
- (D) $x \geq -9$

- 3 Suppose $x^3 - 3x^2 + a \equiv (x - 2)Q(x) + 1$, where $Q(x)$ is a polynomial.
The correct value of a is:
- (A) -2
- (B) 2
- (C) 1
- (D) 5
- 4 A primitive function of $\frac{1}{1 + 4x^2}$ is:
- (A) $\frac{1}{4}\tan^{-1}(2x) + C$
- (B) $\frac{1}{2}\tan^{-1}(2x) + C$
- (C) $\frac{1}{4}\tan^{-1}\left(\frac{x}{2}\right) + C$
- (D) $\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + C$
- 5 If $\frac{dP}{dt} = 0.2(P - 10)$ and $P = 30$ when $t = 0$, which of the following is an expression for P ?
- (A) $P = 10 + 20e^{0.2t}$
- (B) $P = 20 + 10e^{0.2t}$
- (C) $P = 20 + 30e^{0.2t}$
- (D) $P = 30 + 20e^{0.2t}$

6 The gradient of the tangent to the curve $y = \tan^{-1}(\sin x)$ at $x = 0$ is:

(A) undefined

(B) 0

(C) 1

(D) -1

7 The displacement x of a particle at time t is given by:

$$x = 5 \sin 2t + 12 \cos 2t .$$

What is the amplitude of the particle?

(A) 12

(B) 13

(C) 17

(D) 26

8 Choose the correct value of $\lim_{x \rightarrow 0} \frac{\cos(2x) - 1}{x^2}$.

(A) -2

(B) -1

(C) 1

(D) 2

- 9 The rise and fall of the tide is assumed to be simple harmonic, with the time between low and high tide being six hours.

The water depth at a harbour entrance at high and low tides are 14 metres and 10 metres respectively.

If t is the number of hours after low tide, and y is the water depth in metres, which equation models this information?

(A) $y = 12 - 2\cos(6t)$

(B) $y = 12 - 2\cos\left(\frac{\pi t}{6}\right)$

(C) $y = 12 + 2\cos\left(\frac{\pi t}{6}\right)$

(D) $y = 12 + 2\cos(6t)$

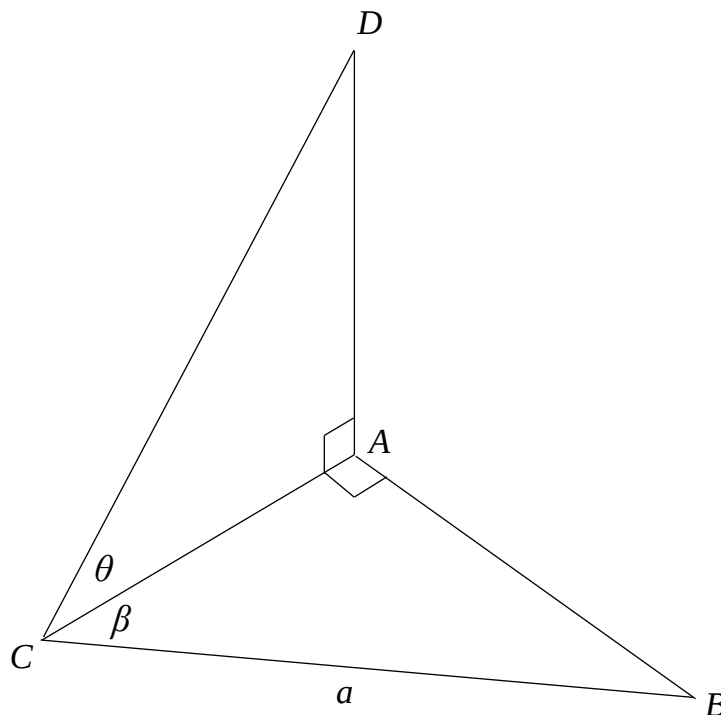
- 10 In the figure, ABC is a triangle on a horizontal plane and AD is a vertical flag pole. If $BC = a$, which of the following expressions is equal to AD ?

(A) $a\sin(\beta + \theta)$

(B) $a\cos\beta\sin\theta$

(C) $a\sin\beta\tan\theta$

(D) $a\cos\beta\tan\theta$



Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.
In Questions 11–14, you should include relevant mathematical reasoning and/ or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

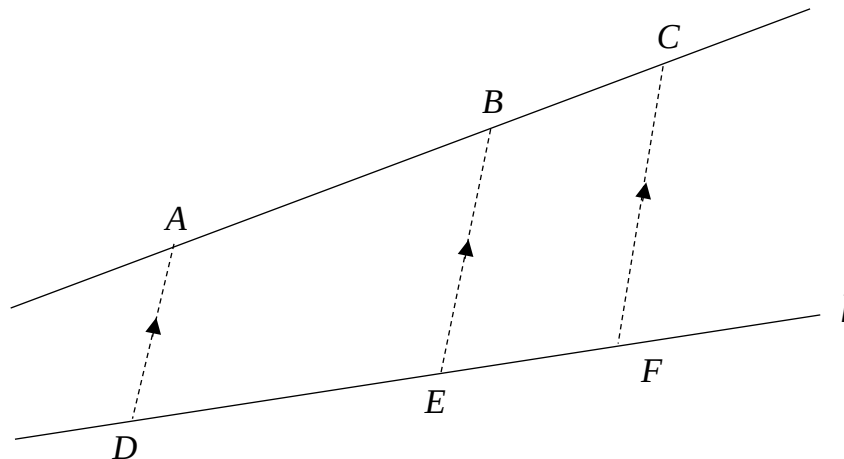
(a) Evaluate $\int_0^1 \frac{dx}{\sqrt{2-x^2}}$. 2

(b) Solve $\frac{2}{3-x} \leq 1$. 3

(c) A is the point $(-1, 1)$, B is the point $(3, 5)$ and C is the point (x, y) .

(i) Find the coordinates of the point C which divides the interval joining AB **externally** in the ratio 3:1. 2

(ii) Parallel lines are drawn through A , B and C and intersect with the line l at the points D , E and F respectively, as shown in the diagram.
What is the ratio of $DE : EF$? 1



NOT TO SCALE

Question 11 continues on the following page.

- (d) An oil leak spreads on the ground in the shape of a circle. The radius of the circle increases from 0 cm, at a constant rate of 5 cm s^{-1} .
At what rate is the area of the circle increasing when the radius is 10 cm?
Express your answer in exact form.

3

- (e) The rate at which a body cools in air is proportional to the difference between its temperature (T) and the constant temperature of the surrounding air (S).

As a result, it follows that:

$$T = S + Be^{kt},$$

where t is the time in hours and B and k are constants.

A metal cake tin has a temperature of 180°C when removed from an oven.

In a room, the temperature of the surrounding air is 20°C .

The cake tin takes 10 minutes to cool to 60°C .

- (i) Show that $k = \frac{\ln(0.25)}{10}$.

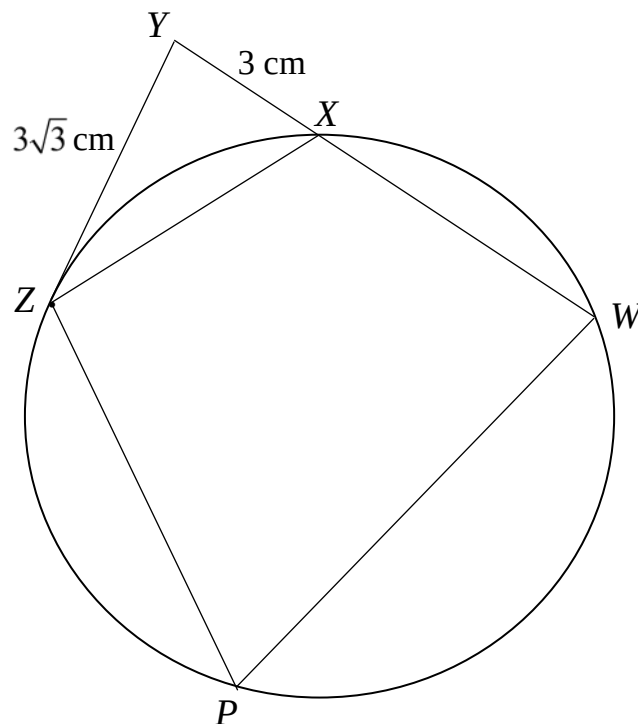
2

- (ii) Hence, find the time it takes for the cake tin to cool to 40°C when removed from the oven.

2

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Use the substitution $u = \sqrt{x}$ to find $\int \frac{dx}{x + \sqrt{x}}$, $x > 0$. 3
- (b) One of the roots of the equation $x^3 - kx^2 + 1 = 0$ is the sum of the other two roots.
- (i) Show that $x = \frac{k}{2}$ is a root of the equation. 2
- (ii) Find the value of k . 1
- (c) In the diagram, the points P , W , X and Z lie on a circle and WX produced meets the tangent from Z at the point Y . It is known that $XY = 3$ cm and $YZ = 3\sqrt{3}$ cm.



NOT TO SCALE

Copy the diagram onto your paper.

- (i) Find the length of XW , giving reasons. 2
- (ii) If XZ is the diameter of a smaller circle passing through X , Y and Z , find the size of $\angle WPZ$, giving reasons. 3

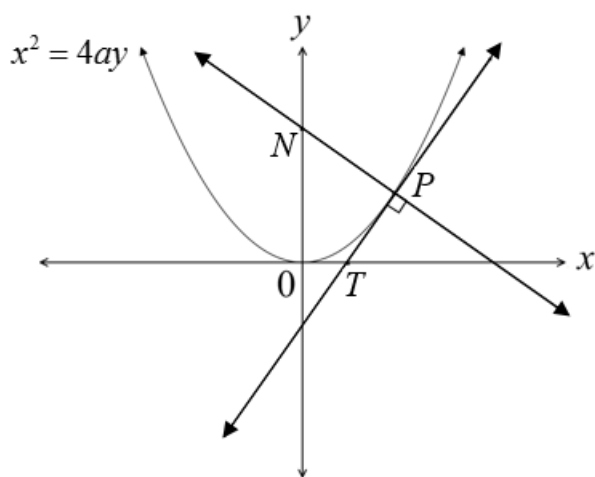
Question 12 continues on the following page.

- (d) Consider the point $P(2ap, ap^2)$ which lies on the parabola $x^2 = 4ay$.
 From the point P , the tangent to the parabola meets the x -axis at T .
 The normal to the parabola at P meets the y -axis at N .

- (i) With the help of the Reference Sheet, show that the coordinates of the points T and N are:

$$T(ap, 0) \quad \text{and} \quad N(0, ap^2 + 2a). \quad 2$$

- (ii) Find the locus of the midpoint M , of TN . 2



NOT TO SCALE

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) Consider the equation $\tan^{-1} x - \frac{x}{2} = 0$.

Using $x = 2$ as the first approximation to one of the roots, use one application of Newton's method to find a better approximation to this root, correct to 2 decimal places. **3**

(b) Find the exact volume of the solid obtained by rotating $y = \cos^{-1} x$ about the y -axis between $y = 0$ and $y = \pi$. **3**

(c) Consider the function $f(x) = 2\sin^{-1}\sqrt{x} - \sin^{-1}(2x-1)$.

(i) Find the domain of $f(x)$. **1**

(ii) Show that $f'(x) = 0$. **2**

(iii) Sketch the graph of $y = f(x)$. **1**

(d) A particle moves in a straight line such that its velocity v m/s is given by $v = 2\sqrt{4x-1}$ when it is x metres from the origin. If $x = \frac{1}{4}$ when $t = 0$, find the:

(i) equation for $\ddot{x}$. **2**

(ii) equation for the displacement (x) in terms of time (t). **3**

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) A particle moves in a straight line such that its acceleration $\ddot{x}$ m/s² is given by $\ddot{x} = 12 - 4x$ when it is x metres from the origin.

Initially, the particle is positioned at $x = 6$ with velocity $v = -2\sqrt{7}$ m/s.

- (i) Explain why the particle moves in simple harmonic motion and state the value of n and the centre of motion. 2
- (ii) Hence show that $v^2 = 28 + 24x - 4x^2$. 2
- (iii) Over what range of x -values is the particle moving? 1
- (iv) Find the maximum speed of the particle. 1
- (b) Use the principle of mathematical induction to prove that $2^{3n} - 3^n$ is divisible by 5 for all integers $n \geq 1$. 3

Question 14 continues on the following page.

- (c) Jon Snow fires arrows with an initial velocity V m/s at an angle of 60° to the horizontal. The path of any arrow from the origin is given by the displacement functions below, where t is the time in seconds and g is the acceleration due to gravity in m/s^2 :

$$x = \frac{Vt}{2}$$

(DO NOT PROVE THESE)

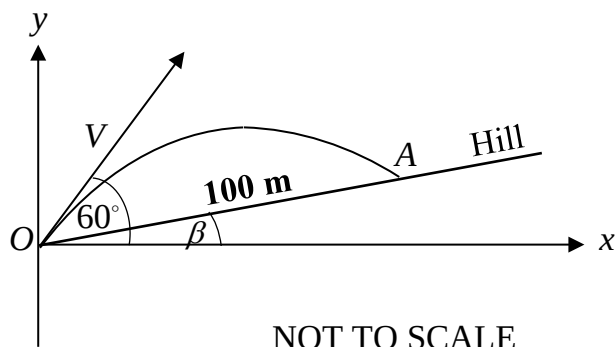
$$y = -\frac{1}{2}gt^2 + \frac{Vt\sqrt{3}}{2}$$

- (i) Show that the Cartesian equation of an arrow's path is: $y = \sqrt{3}x - \frac{2gx^2}{V^2}$. 1

- (ii) Jon stood at the bottom of a hill inclined at an angle β to the horizontal. When he fired an arrow, it landed 100 metres up the hill at point A.

Use the result of part (i) to show that:

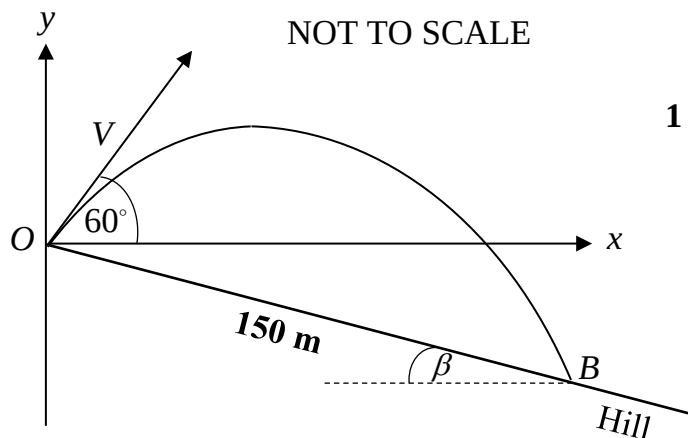
$$\tan \beta = \sqrt{3} - \frac{200g \cos \beta}{V^2}$$



- (iii) Jon climbs the hill. Jon turns around and is now standing at the top of the same hill, looking down the hill. When he shoots an arrow at the same velocity and angle of projection, the arrow lands 150 metres down the hill, at point B.

Show that:

$$\tan \beta = \frac{300g \cos \beta}{V^2} - \sqrt{3}$$



- (iv) Hence, show that $\beta = \tan^{-1}\left(\frac{\sqrt{3}}{5}\right)$. 2

End of Exam

① Reflex $\angle AOC = 2 \times \angle ADC$
 $= 260^\circ$

($\angle$ at centre is twice $\angle$ at circumference)

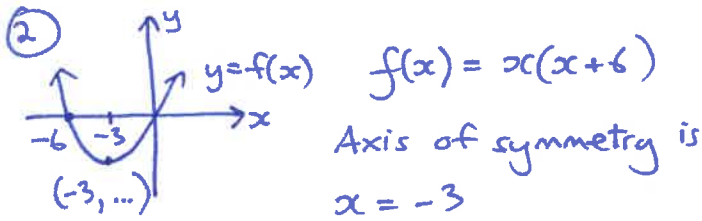
$\angle OCB = 35^\circ$ (base $\angle$ s isosceles $\triangle$ equal)

$\angle BOC = 180^\circ - 2 \times 35^\circ$ ($\angle$ sum of $\triangle$)
 $= 110^\circ$

$\alpha = 260^\circ - 110^\circ$

$\therefore \alpha = 150$

$\therefore$ C



$\therefore$ Largest domain for $f^{-1}(x)$ is
 $x \geq -3$ or $x \leq -3$ $\therefore$ B

③ If $P(x) = x^3 - 3x^2 + a$,
 then $P(2) = 1$, since
 $P(x) = (x-2)Q(x) + 1$
 Hence $2^3 - 3(2)^2 + a = 1$
 $8 - 12 + a = 1$
 $\therefore a = 5$ $\therefore$ D

④ $\int \frac{1}{1+4x^2} dx$

$= \int \frac{1}{4(\frac{1}{4} + x^2)} dx$

$= \frac{1}{4} \int \frac{1}{(\frac{1}{2})^2 + x^2} dx$

$= \frac{1}{4} \times \frac{1}{\frac{1}{2}} \tan^{-1}\left(\frac{x}{\frac{1}{2}}\right) + C$

$= \frac{1}{2} \tan^{-1}(2x) + C$

$\therefore$ B

⑤ If $\frac{dP}{dt} = k(P-B)$, then

$P = B + Ae^{kt}$

Given $\frac{dP}{dt} = 0.2(P-10)$

then $P = 10 + Ae^{0.2t}$

$\therefore$ A

[since when $30 = 10 + Ae^0$,
 then $A = 20$]

⑥ If $f(x) = \tan^{-1}(\sin x)$

$f'(x) = \frac{1}{1+\sin^2 x} \times \cos x$

Now $f'(0) = \frac{\cos(0)}{1+\sin^2(0)} = \frac{1}{1+0} = 1$

$\therefore$ C

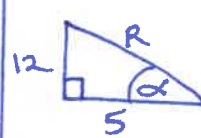
⑦ Using $R \sin(2t + \alpha)$

$= R \sin(2t) \cos \alpha + R \cos(2t) \sin \alpha$

$R \cos \alpha = 5$ and $R \sin \alpha = 12$

$\cos \alpha = \frac{5}{R}$

$\sin \alpha = \frac{12}{R}$



$R = \sqrt{5^2 + 12^2}$

$\therefore R = 13$

$\therefore$ B

⑧ $\lim_{x \rightarrow 0} \frac{\cos(2x) - 1}{x^2}$

$= \lim_{x \rightarrow 0} \frac{1 - 2\sin^2 x - 1}{x^2}$

$= -2 \lim_{x \rightarrow 0} \frac{\sin x}{x} \lim_{x \rightarrow 0} \frac{\sin x}{x}$

$= -2 \times 1 \times 1$

$= -2$

$\therefore$ A



Amplitude = 2

Period = 12 hours

$$\therefore \frac{2\pi}{n} = 12$$

$$\therefore n = \frac{\pi}{6}$$

$$y = b + a \cos(nt)$$

$$\Rightarrow y = 12 - 2 \cos\left(\frac{\pi}{6}t\right) \therefore \underline{B}$$

10) In $\triangle ABC$: $\cos \beta = \frac{AC}{a}$

$$\therefore AC = a \cos \beta$$

In $\triangle ACD$: $\tan \theta = \frac{AD}{AC}$

$$\therefore \tan \theta = \frac{AD}{a \cos \beta}$$

$$\therefore AD = a \cos \beta \tan \theta \therefore \underline{D}$$

Question 11

a) $\int_0^1 \frac{1}{\sqrt{(\sqrt{2})^2 - x^2}} dx$

$$= \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_0^1$$

$$= \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) - \sin^{-1} 0$$

$$= \frac{\pi}{4} - 0$$

$$= \frac{\pi}{4}$$

(2 marks)

b) $\frac{2}{3-x} \leq 1, x \neq 3$ (2)

$$\frac{2(3-x)^2}{3-x} \leq 1(3-x^2)$$

$$2(3-x) \leq (3-x)^2$$

$$2(3-x) - (3-x)^2 \leq 0$$

$$(3-x)[2 - (3-x)] \leq 0$$

$$(3-x)(x-1) \leq 0$$

$$\therefore x \leq 1$$

or

$$x > 3$$

$\frac{1}{2}$

$\frac{1}{2}$



(3 marks)

c) i) Let ratio be $m:n$
 $= 3:-1$

If $(x_1, y_1) = (-1, 1)$

and $(x_2, y_2) = (3, 5)$

Use $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$

$$C = \left(\frac{3(3) - 1(-1)}{3 + -1}, \frac{3(5) - 1(1)}{3 + -1} \right)$$

$$C = (5, 7) \quad (2 \text{ marks})$$

Deduct 1 mark for sign error

ii) Ratio of intercepts on parallel lines are equal

$$\therefore DE:EF = AB:BC$$

If C divides AB externally in the ratio 3:1, then
 $AB:BC = 2:1$

$$\therefore DE:EF = 2:1$$

(1 mark)

d) $A = \pi r^2$

$$\frac{dA}{dr} = 2\pi r$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$= 2\pi r \times 5$$

$\therefore$ when $r = 10$,

$$\frac{dA}{dt} = 100\pi \text{ cm}^2/\text{s}$$

(3 marks)

e) i) $T = S + Be^{kt}$

$$180^\circ = 20^\circ + Be^{k \times 0}$$

$$\therefore B = 160^\circ$$

So $T = 20 + 160e^{kt}$

when $t = 10$, $T = 60$

$$60 = 20 + 160e^{10k}$$

$$\frac{40}{160} = e^{10k}$$

$$\ln\left(\frac{1}{4}\right) = 10k$$

$$\therefore k = \frac{\ln(0.25)}{10}$$

(2 marks)

ii) $40 = 20 + 160e^{kt}$

$$\frac{40 - 20}{160} = e^{kt}$$

$$\frac{1}{8} = e^{kt}$$

$$\ln\left(\frac{1}{8}\right) = kt$$

$$t = \frac{\ln\left(\frac{1}{8}\right)}{k}$$

$$t = \ln\left(\frac{1}{8}\right)$$

$$\left[\frac{\ln(0.25)}{10} \right]$$

$$t = 15$$

$\therefore$ It takes 15 minutes to cool to 40°C

(2 marks)

(3)

Question 12

a) If $u = \sqrt{x}$, $\frac{du}{dx} = \frac{1}{2}x^{-\frac{1}{2}}$

$$du = \frac{dx}{2\sqrt{x}}$$

$$dx = 2\sqrt{x} du$$

$$\therefore dx = 2u du$$

$$\int \frac{dx}{x + \sqrt{x}}$$

$$= \int \frac{2u}{u^2 + u} du$$

$$= \int \frac{2u}{u(u+1)} du$$

$$= \int \frac{2}{u+1} du$$

$$= 2\ln(u+1) + c$$

$$= 2\ln(\sqrt{x} + 1) + c$$

(3 marks)

b) i) Let roots be $\alpha, \beta, \alpha + \beta$

$$\text{sum of roots} = -\frac{b}{a}$$

$$\alpha + \beta + \alpha + \beta = -\frac{k}{1}$$

$$2(\alpha + \beta) = -k$$

$$\alpha + \beta = -\frac{k}{2}$$

$$\therefore \text{one of the roots is } x = -\frac{k}{2}$$

ii) $\left(\frac{k}{2}\right)^3 - k\left(\frac{k}{2}\right)^2 + 1 = 0$

$$\frac{k^3}{8} - \frac{k^3}{4} = -1$$

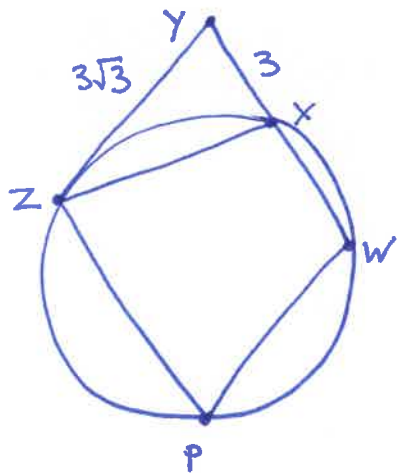
$$\frac{k^3 - 2k^3}{8} = -1$$

$$-k^3 = -8$$

$$\therefore k = 2$$

(1 mark)

c)



$$(i) YZ^2 = YW \times YX$$

(square of tangent equals product of intercepts of secant) ✓

$$\text{Let } XW = x$$

$$(3\sqrt{3})^2 = (3+x) \times 3$$

$$27 = 9 + 3x$$

$$x = 6$$

$$\therefore XW = 6 \text{ cm}$$

(2 marks) ✓

$$(ii) \angle ZYX = 90^\circ$$

($\angle$ in a semi-circle equals 90°) ✓

$$\text{In } \triangle XYZ: \tan \angle YXZ = \frac{3\sqrt{3}}{3}$$

$$\therefore \angle YXZ = 60^\circ \quad \checkmark$$

$$\therefore \angle WPZ = 60^\circ$$

(exterior $\angle$ of cyclic quadrilateral equals interior opposite $\angle$) ✓

(3 marks)

$$d) \quad (i) \text{ Tangent at } P: y = px - ap^2$$

$$\text{At } T, \text{ where } y=0: 0 = px - ap^2$$

$$px = ap^2$$

$$x = \frac{ap^2}{p}$$

$$x = ap$$

$$\therefore T \text{ is } (ap, 0)$$

(1 mark)

$$\text{Normal at } P: x + py = ap^3 + 2ap \quad (4)$$

At N, where $x=0$:

$$0 + py = ap^3 + 2ap$$

$$y = \frac{ap^3 + 2ap}{p}$$

$$y = ap^2 + 2a$$

$$\therefore N \text{ is } (0, ap^2 + 2a)$$

(1 mark)

(ii) Midpoint (M) of TN

$$= \left(\frac{ap+0}{2}, \frac{0+ap^2+2a}{2} \right)$$

$$\text{i.e. } x = \frac{ap}{2} \quad \text{and} \quad y = \frac{ap^2+2a}{2} \quad (2)$$

$$\therefore p = \frac{2x}{a} \quad \dots (1) \quad \checkmark$$

sub. eqn (1) into (2):

$$y = \frac{a\left(\frac{2x}{a}\right)^2 + 2a}{2}$$

$$y = \frac{\frac{4x^2}{a} + 2a}{2}$$

$$\therefore y = \frac{2x^2}{a} + a \quad \checkmark$$

(2 marks)

Question 13

a) If $f(x) = \tan^{-1}x - \frac{x}{2}$

then $f'(x) = \frac{1}{1+x^2} - \frac{1}{2}$ ✓

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

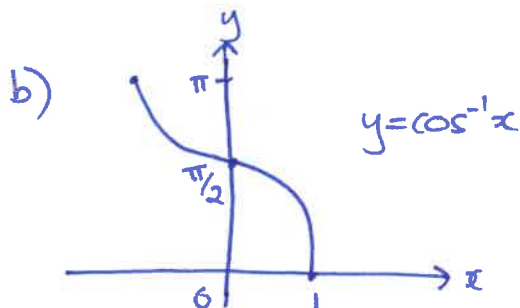
Using $x_1 = 2$:

$$x_2 = 2 - \left[\frac{\tan^{-1}(2) - \frac{2}{2}}{\frac{1}{1+2^2} - \frac{1}{2}} \right] \quad \checkmark$$

$$x_2 = 2.357\dots$$

$$\therefore x_2 = 2.36 \text{ (2 d.p.)} \quad \checkmark$$

(3 marks)



Making x the subject:

$$x = \cos y$$

$$V = 2\pi \int_0^{\pi/2} \cos^2 y \, dy \quad \checkmark$$

$$V = 2\pi \int_0^{\pi/2} \frac{1}{2} (\cos 2y + 1) \, dy$$

$$V = \pi \left[\frac{\sin 2y}{2} + y \right]_0^{\pi/2} \quad \checkmark$$

$$V = \pi \left[\frac{\sin(2 \times \pi/2)}{2} + \frac{\pi}{2} - \left(\frac{\sin(0)}{2} + 0 \right) \right]$$

$$V = \pi \left[0 + \frac{\pi}{2} - 0 \right]$$

$$\therefore V = \frac{\pi^2}{2} \text{ units}^3 \quad \checkmark$$

(3 marks)

c) $f(x) = 2\sin^{-1}\sqrt{x} - \sin^{-1}(2x-1)$ (5)

i) $x \geq 0$ and $0 \leq \sqrt{x} \leq 1 \therefore 0 \leq x \leq 1$

and $-1 \leq 2x-1 \leq 1$
 $0 \leq 2x \leq 2$

$$\therefore 0 \leq x \leq 1 \quad \checkmark$$

(1 mark)

ii)

$$f'(x) = \frac{2}{\sqrt{1-(\sqrt{x})^2}} \times \frac{1}{2} x^{-\frac{1}{2}} - \frac{1}{\sqrt{1-(2x-1)^2}} \times 2 \quad \checkmark$$

$$= \frac{1}{\sqrt{x}\sqrt{1-x}} - \frac{2}{\sqrt{1-(4x^2-4x+1)}}$$

$$= \frac{1}{\sqrt{x}\sqrt{1-x}} - \frac{2}{\sqrt{4x(x-1)}} \quad \frac{1}{2}$$

$$= \frac{1}{\sqrt{x}\sqrt{1-x}} - \frac{2}{2\sqrt{x}\sqrt{x-1}} \quad \frac{1}{2}$$

$$= \frac{1}{\sqrt{x}\sqrt{1-x}} - \frac{1}{\sqrt{x}\sqrt{x-1}}$$

$$= 0$$

(2 marks)

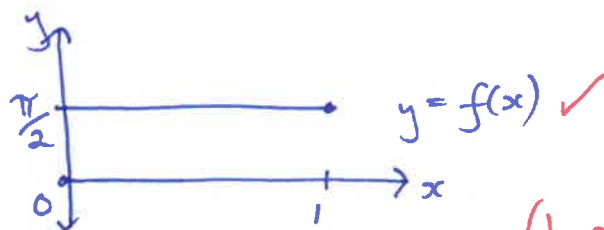
iii) when $x=0$:

$$f(0) = 2\sin^{-1}\sqrt{0} - \sin^{-1}[2(0)-1]$$

$$= 2 \times 0 - \sin^{-1}(-1)$$

$$= 0 - (-\pi/2)$$

$$= \frac{\pi}{2}$$



(1 mark)

Deduct $\frac{1}{2}$ mk for incorrect domain.

(6)

$$d) i) \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$\ddot{x} = \frac{d}{dx} \left[\frac{1}{2} (2\sqrt{4x-1})^2 \right] \quad \checkmark$$

$$\ddot{x} = \frac{d}{dx} [2(4x-1)]$$

$$\therefore \ddot{x} = 8 \text{ms}^{-2}$$

✓
(2 marks)

$$ii) \frac{dx}{dt} = 2\sqrt{4x-1}$$

$$\frac{dt}{dx} = \frac{1}{2\sqrt{4x-1}} \quad \frac{1}{2}$$

$$t = \int \frac{(4x-1)^{-\frac{1}{2}}}{2} dx \quad \frac{1}{2}$$

$$t = \frac{(4x-1)^{\frac{1}{2}}}{2 \times \frac{1}{2} \times 4} + C$$

$$t = \frac{\sqrt{4x-1}}{4} + C \quad \checkmark$$

$$\text{when } t=0, x = \frac{1}{4}$$

$$0 = \frac{\sqrt{4(\frac{1}{4})-1}}{4} + C$$

$$\therefore C = 0 \quad \frac{1}{2}$$

$$\therefore t = \frac{\sqrt{4x-1}}{4}$$

$$4t = \sqrt{4x-1}$$

$$16t^2 = 4x-1$$

$$4x = 16t^2 + 1$$

$$\therefore x = \frac{16t^2 + 1}{4} \quad \frac{1}{2}$$

(3 marks)

Question 14

a) i) $\ddot{x} = 12 - 4x$

$$\ddot{x} = -4(x-3)$$

$$\ddot{x} = -(2)^2(x-3)$$

since $\ddot{x}$ is of the form $\ddot{x} = -n^2(x-b)$

the particle moves in SHM where

$n=2$ and centre is $x=3$

ii) $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$$12 - 4x = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$\int 12 - 4x \, dx = \frac{1}{2} v^2$$

$$12x - \frac{4x^2}{2} + C = \frac{1}{2} v^2$$

when $x=6$ $v = -2\sqrt{5}$

$$12(6) - 2(6)^2 + C = \frac{1}{2} (-2\sqrt{5})^2$$

$$72 - 72 + C = \frac{1}{2} (28)$$

$$\therefore C = 14$$

$$\frac{1}{2} v^2 = 12x - 2x^2 + 14$$

$$\therefore v^2 = 28 + 24x - 4x^2$$

(as required)

iii) Solve $28 + 24x - 4x^2 = 0$

$$4x^2 - 24x - 28 = 0$$

$$4(x^2 - 6x - 7) = 0$$

$$4(x-7)(x+1) = 0$$

$$\therefore x=7 \text{ or } x=-1$$

so particle moves in the

range $-1 \leq x \leq 7$

iv) Max speed occurs at $x=3$ (centre of motion)

$$v^2 = 28 + 24(3) - 4(3)^2$$

$$v^2 = 64$$

$$\therefore \text{max speed} = \sqrt{64} = 8 \text{ m/s}$$

7

b) Step ① Prove true for $n=1$

$$2^{3(1)} - 3^1 = 8 - 3 = 5$$

which is divisible by 5

$\therefore$ it is true for $n=1$

Step ② Assume it is true for $n=k$

i.e. $2^{3k} - 3^k = 5M$ ($M \in \mathbb{Z}$)

$$\therefore 2^{3k} = 5M + 3^k$$

Step ③ Prove it is true for $n=k+1$

i.e. RTP $2^{3(k+1)} - 3^{k+1} = 5Q$ ($Q \in \mathbb{Z}$)

$$\text{LHS} = 2^{3k+3} - 3^{k+1}$$

$$= 2^{3k} \times 2^3 - 3^k \times 3^1$$

$$= (5M + 3^k) 8 - 3^k \times 3 \dots \text{using } *$$

$$= 40M + 8(3^k) - 3(3^k)$$

$$= 40M + 5(3^k)$$

$$= 5(8M + 3^k)$$

which is divisible by 5

Hence it is proved true by Mathematical

Induction.

Alternate Solution: (3 marks)

Step ② $2^{3k} - 3^k = 5M$

Step ③ LHS = $2^{3k} \times 2^3 - 3 \times 3^k$

$$= 2^{3k}(8) - 3(2^{3k} - 5M)$$

$$= 8(2^{3k}) - 3(2^{3k}) + 15M$$

$$= 5(2^{3k}) + 15M$$

$$= 5(2^{3k} + 3M)$$

which is divisible by 5

c) i) From $x = \frac{vt}{2}$, $t = \frac{2x}{v}$
 sub. into $y = -\frac{1}{2}gt^2 + \frac{vt\sqrt{3}}{2}$

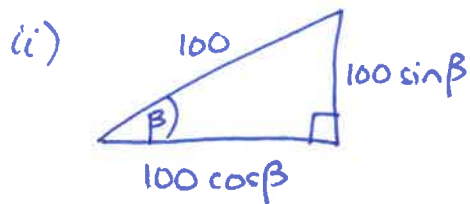
$$\Rightarrow y = -\frac{1}{2}g\left(\frac{2x}{v}\right)^2 + v\left(\frac{2x}{v}\right)\frac{\sqrt{3}}{2}$$

$$y = -\frac{2gx^2}{v^2} + x\sqrt{3}$$

i.e. $y = \sqrt{3}x - \frac{2gx^2}{v^2}$

(As required)

(1 mark)



sub. $x = 100 \cos \beta$
 and $y = 100 \sin \beta$ into

$$y = \sqrt{3}x - \frac{2gx^2}{v^2}$$

$$100 \sin \beta = \sqrt{3}(100 \cos \beta) - \frac{2g(100 \cos \beta)^2}{v^2}$$

$$100 \sin \beta = 100\sqrt{3} \cos \beta - \frac{20000g \cos^2 \beta}{v^2}$$

Divide both sides by $100 \cos \beta$:

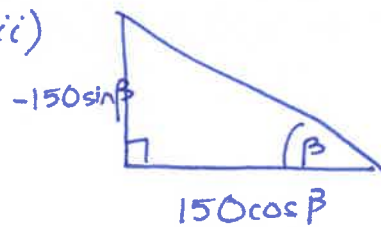
$$\frac{100 \sin \beta}{100 \cos \beta} = \frac{100\sqrt{3} \cos \beta}{100 \cos \beta} - \frac{20000g \cos^2 \beta}{v^2 100 \cos \beta}$$

$$\therefore \tan \beta = \sqrt{3} - \frac{200g \cos \beta}{v^2} \dots (1)$$

(as required)

(2 marks)

(iii)



sub. $x = 150 \cos \beta$
 and $y = 150 \sin \beta$ into

$$y = \sqrt{3}x - \frac{2gx^2}{v^2}$$

$$150 \sin \beta = \sqrt{3}(150 \cos \beta) - \frac{2g(150 \cos \beta)^2}{v^2}$$

$$150 \sin \beta = 150\sqrt{3} \cos \beta - \frac{4500g \cos^2 \beta}{v^2}$$

Divide both sides by $150 \cos \beta$:

$$\frac{150 \sin \beta}{150 \cos \beta} = \frac{150\sqrt{3} \cos \beta}{150 \cos \beta} - \frac{4500g \cos^2 \beta}{v^2 150 \cos \beta}$$

$$-\tan \beta = \sqrt{3} - \frac{300g \cos \beta}{v^2}$$

i.e. $\tan \beta = \frac{300g \cos \beta}{v^2} - \sqrt{3} \dots (2)$
 (as required) (1 mark)

(iv)

① $\times 3$: $3 \tan \beta = 3\sqrt{3} - \frac{600g \cos \beta}{v^2} \dots (3)$

② $\times 2$: $2 \tan \beta = \frac{600g \cos \beta}{v^2} - 2\sqrt{3} \dots (4)$

③ + ④: $5 \tan \beta = 3\sqrt{3} + -2\sqrt{3}$

$$5 \tan \beta = \sqrt{3}$$

$$\tan \beta = \frac{\sqrt{3}}{5}$$

$$\therefore \beta = \tan^{-1}\left(\frac{\sqrt{3}}{5}\right)$$

(as required)

(1 mark)
 [Award 1 mk for some progress in right direction]
 P.T.O. for alternate solution...

END OF EXAM

Alternate solution to iii)

Rearranging eqn (1):

$$\frac{g \cos \beta}{v^2} = \frac{\tan \beta - \sqrt{3}}{-200}$$

sub. into eqn (2):

$$\tan \beta = 300 \left(\frac{\tan \beta - \sqrt{3}}{-200} \right) - \sqrt{3}$$

$$200 \tan \beta = -300 (\tan \beta - \sqrt{3}) - 200 \sqrt{3}$$

$$200 \tan \beta = -300 \tan \beta + 300 \sqrt{3} - 200 \sqrt{3}$$

$$500 \tan \beta = 100 \sqrt{3}$$

$$\tan \beta = \frac{100 \sqrt{3}}{500}$$

$$\tan \beta = \frac{\sqrt{3}}{5}$$

$$\therefore \beta = \tan^{-1} \left(\frac{\sqrt{3}}{5} \right)$$